

11. When documenting the current process, which of the following tools can be helpful?
 A) Organizational charts. B) Gantt charts.
 C) Flow diagrams and cause-and-effect diagrams. D) PERT charts.
12. Process benchmarking is also known as:
 A) Competitive benchmarking. B) Internal benchmarking.
 C) Functional or generic benchmarking. D) Strategic benchmarking.
13. A negative gap in a benchmarking study indicates that:
 A) The internal process is better than the external process. B) The external process is significantly better than the internal process.
 C) The processes are approximately equal. D) The benchmarking study was unsuccessful.
14. The two types of information that benchmarking studies look for are:
 A) Financial data and market share. B) A description of how best-in-class processes are practiced and their measurable results.
 C) Customer complaints and employee satisfaction. D) Historical data and future projections.
15. Who coined the phrases "vital few" and "useful many"?
 A) Alfredo Pareto B) W. Edwards Deming
 C) Dr. Kaoru Ishikawa D) Dr. Joseph Juran
16. The horizontal scale of a Pareto diagram is:
 A) Numerical B) Time-based
 C) Categorical D) Continuous
17. A process flow diagram is useful for:
 A) Determining the statistical mean of a process B) Visualizing the entire system and identifying potential trouble spots
 C) Calculating the standard deviation of a sample D) Identifying the "vital few" problems in a process
18. Who developed the Cause-and-Effect diagram?
 A) Alfredo Pareto B) Dr. Kaoru Ishikawa
 C) Dr. Joseph Juran D) W. Edwards Deming
19. The main purpose of check sheets is to:
 A) Generate new ideas through brainstorming B) Rank data classifications in descending order
 C) Ensure that data is collected carefully and accurately D) Graphically estimate process capability
20. What is the first step when processing unorganized data to create a histogram?
 A) Compute the average B) Tabulate the frequency of each value
 C) Establish an array of raw numerical data in ascending or descending order D) Group the data into cells

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Show a process model diagram. Discuss are the five ways to improve a process.	08	L*1	1	1
	b) Discus the three Levels of Quality.	08	L1	1	1
2.	a) What are muda, muri and mura? Explain the types of muda.	08	L2	2	1
	b) Explain 5S with suitable examples.	08	L2	2	1
3.	a) List the differences between quality control and quality management.	08	L2	1	1
	b) Discuss the building blocks of Lean.	08	L2	2	1
Unit – II					
4.	a) Define Benchmarking with a neat diagram. What are the two benchmarking key elements? List any five reasons for Benchmarking.	08	L2	3	1

b)	Show the structure of QFD's House of Quality. Explain the first two steps that can be used to build the house of quality.	08	L2	3	1
5. a)	What is FMEA? List the benefits of design and process FMEA. Give the four stages of FMEA.	08	L2	4	1
b)	List the key Six Sigma concepts. Explain briefly the following concepts				
i.	The Six Sigma Quality Level				
ii.	Six Sigma versus Three Sigma	08	L2	4	1
6. a)	What is Quality Function Deployment (QFD)? Explain the Benefits of QFD.	08	L2	5	1
b)	How do you describe DMAIC? Explain the Phase-I of DMAIC.	08	L2	5	1
Unit – III					
7. a)	Discuss Pareto diagram with suitable examples.	08	L2	5	1
b)	What are the various reasons for implementing a quality system? Write a note on ISO 9000 Series of Standards.	08	L1	5	1
8. a)	Explain with suitable examples how cause-and-effect diagram can be used to improving product and service quality.	08	L2	5	1
b)	Define and explain the principles of Sustainable Development.	08	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
